Session: B

Paper no: 2

Effect of Force-Rate on Continuous Kinesthetic Force Discrimination

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Introduction

- Conventional haptic force discrimination tasks involve perceptual differentiation between stimulus pairs separated by a brief inter-stimulus interval (ISI).
- It is unclear whether the **Force JND** measured using conventional methods, i.e., with ISI in between stimuli, remains consistent in the case of **Continuous Force Discrimination**.
- Weber Fraction, $\delta = \frac{|F_t - F_r|}{|F_r|}$, where, F_r = Reference Force, F_t = test force

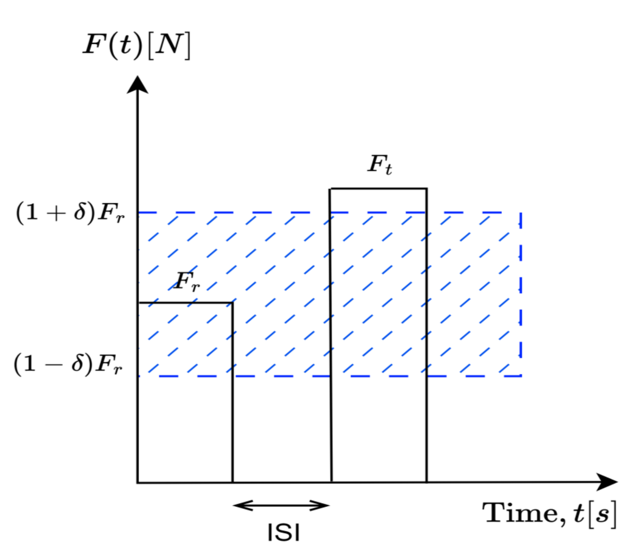

Fig.1: Force discrimination with ISI and corresponding deadzone (dashed area)

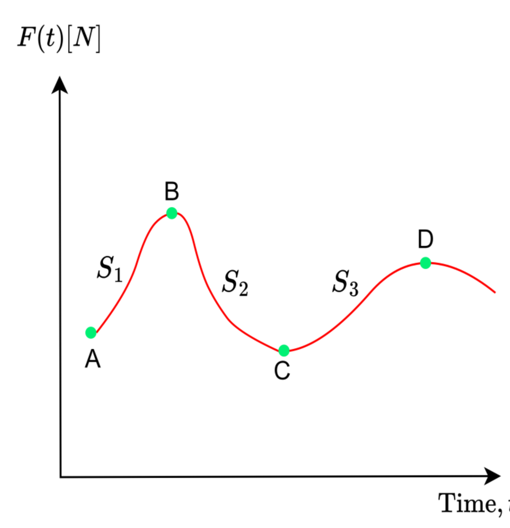

Fig.2: Continuous force stimuli $F(t)$ with varying force-rates S_1, S_2, and S_3

Methodology

- Machine learning classification-based approach
- Paradigm: Participant is exposed to a linearly increasing 1D kinesthetic force stimulus and is asked to react to the change in force.

Setup-

- PHANToM Premium 1.5 HF haptic device
- One-dimensional (1D) haptic force stimulus → +Z-axis

Fig.3: Experimental setup

Stimuli-

- $R\epsilon[1,5]N/s$, $D\epsilon[0.1,.75]N$

Data-Collection-

- Participants: 10 users
- Total trials: 1000 trials per user

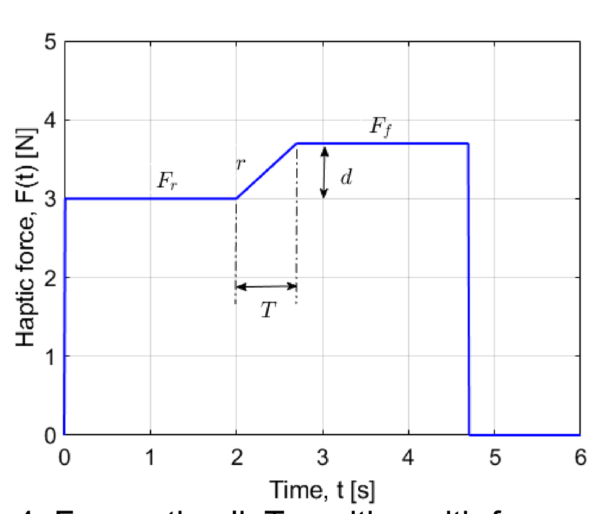

Fig.4: Force stimuli: Transition with force-rate (r)

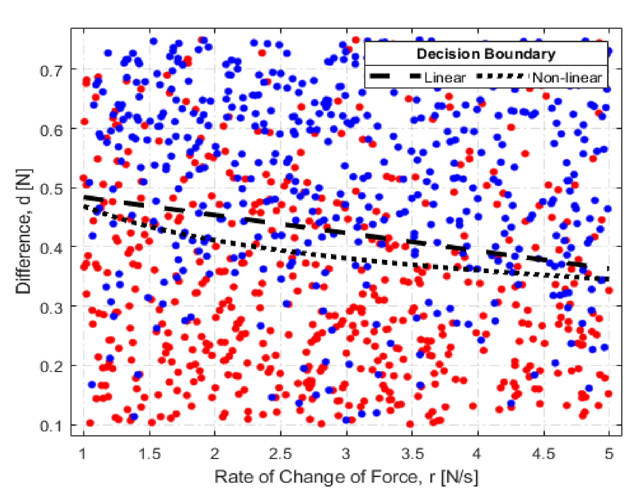

Fig. 5: Scatter plot for one participant

Estimation of Decision Boundary

Classification error: $$E^k = \frac{1}{N}\sum_{i=1}^{N}\left(w_p\frac{(1+y_i)}{2} + w_{np}\frac{(1-y_i)}{2}\right)\mathbf{1}(y_i \neq h^k(r_i,d_i))$$

$$w_p = \frac{N_{np}}{N_p+N_{np}} \quad \text{and} \quad w_{np} = \frac{N_p}{N_p+N_{np}}$$

1. Linear function-based boundary

$$d = mr + c \qquad h^l(r_i,d_i) = \begin{cases} 1 & \text{for } d_i - (mr_i + c) \geq 0 \\ -1 & \text{for } d_i - (mr_i + c) < 0. \end{cases}$$

2. Power function-based boundary

$$d = \beta r^{\alpha} \qquad h^p(r_i,d_i) = \begin{cases} 1 & \text{for } (d_i - \beta r_i^{\alpha}) \geq 0 \\ -1 & \text{for } (d_i - \beta r_i^{\alpha}) < 0 \end{cases}$$

Results and Analysis

- Boundary structures are validated using a Random Forest Classifier
- Mean prediction error- $E^l = 27.94\%$, $E^p = 27.11\%$
- No significant difference in error (t=1.388, p=0.198)
- Two-way ANOVA with factors: *force-rate* and *perceptual boundaries*
  - A significant effect of force-rate on Weber fraction ($F(2,54) = 55.27$, $p < 0.001$)
  - A significant effect of perceptual boundaries on Weber fraction ($F(1,54) = 33.66$, $p < 0.001$)
  - The interaction effect between force-rate and perceptual boundaries is not significant ($F(2,54) = 0.83$, $p = 0.440$)

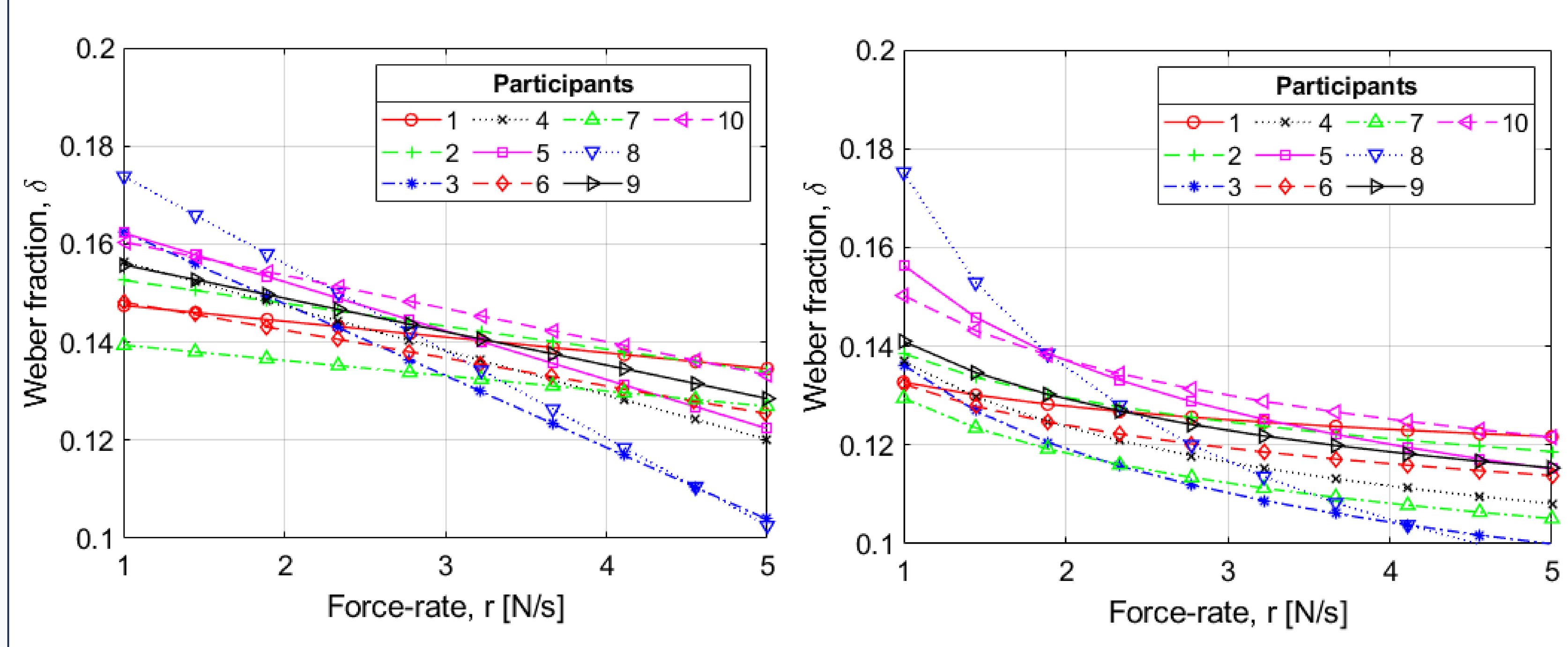

Fig.6: (a) Linear, (b) Power function-based boundaries

Discussion & Conclusion

- Force-rate (r) significantly influences continuous kinesthetic force discrimination.
- Weber fraction (δ) decreases with increasing positive force-rate in the range **[1–5] N/s**, indicating improved sensitivity to force changes.
- The observed variation in δ suggests that haptic force perception is **rate-dependent** rather than constant during continuous interaction.
- The proposed **perceptual boundary models** provide a framework for characterizing continuous haptic force perception.
- These findings may inform future **perceptually adaptive haptic codecs and communication frameworks**, where transmission thresholds or update strategies are adjusted according to force-rate while maintaining perceived interaction fidelity.

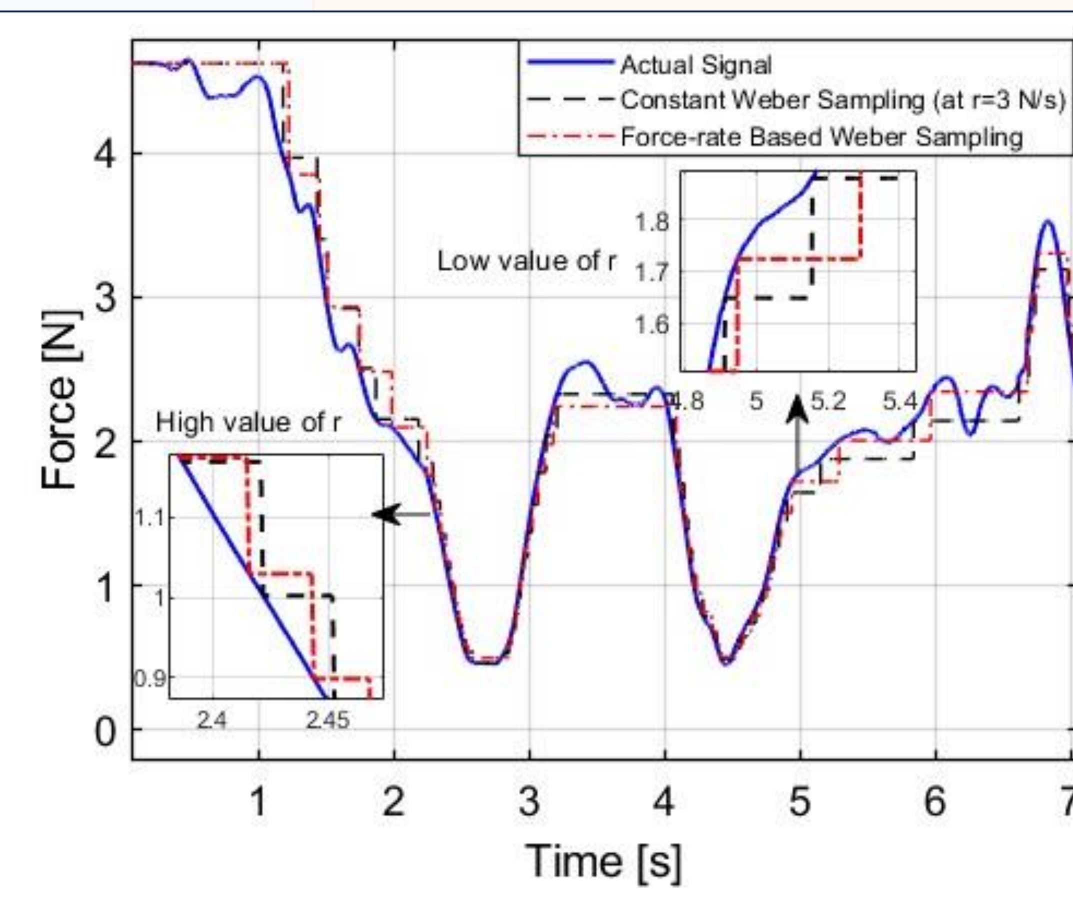

Fig.7: Force signal and reconstructed signals using both Constant Weber fraction (at r = 3 N/s) and force-rate-based Weber fraction sampling for one participant

Meghwanshi, R.P., Bhardwaj, A. & Kumar, H. Effect of force-rate on continuous kinesthetic force discrimination. Exp Brain Res 243, 118 (2025)